Java Access Modifiers (Access Specifiers)

Access Modifiers in Java are used to control the visibility and accessibility of classes, methods, variables, and constructors. They help achieve data security, encapsulation, and better object-oriented design.

Types of Access Modifiers in Java

Java provides four types of access modifiers: public, private, protected, and default (no keyword).

Modifier	Same Class	Same Package	Subclass (Diff Package)	Other Classes
public	Yes	Yes	Yes	Yes
protected	Yes	Yes	Yes	No
default	Yes	Yes	No	No
private	Yes	No	No	No

1. public

Members declared as public are accessible from anywhere in the program.

2. private

Members declared as private are accessible only within the same class. This is widely used to achieve encapsulation.

3. default (no keyword)

When no access modifier is specified, Java applies default access. Members are accessible only within the same package.

4. protected

Protected members are accessible within the same package and also in subclasses even if they are in different packages.

Important Notes

- Top-level classes can only be public or default.
- Variables and methods can use all four access modifiers.
- private is the most restrictive and public is the least restrictive.

Interview Tip

Access modifiers are a core concept of Java OOP. Interviewers often ask questions comparing protected and default or testing visibility rules.